

Suggestions:

- Life Science Laboratory
Cost: \$30 Million
Mass: 20kg
Power Draw: 20W
Science Return: 15
- High-Resolution Camera (Camera) (Buying two)
Cost (for 1): \$20 Million
Mass (for 1): 6kg
Power Draw (for 1): 4W
Science Return (for 1): 5
- Color Stereo Camera (Camera)
Cost: \$15 Million
Mass: 5kg
Power Draw: 3W
Science Return: 5
- Radiation Sensor (Sensor)
Cost: \$10 Million
Mass: 1kg
Power Draw: 1W
Science Return: 1
- Advanced Resistive Exercise Device (ARED)
Cost: \$8 Million
Mass: 5kg
Power Draw: 2W
Science Return: 5

Total Summary For Instruments:

- Total Cost: \$103 Million
- Total Mass: 43kg
- Total Power Draw: 34W
- Total Science Return: 36

Reasons:

The **Life Sciences Laboratory** is an essential part of this mission and I definitely suggest it because the function it serves is that it supports food biology, astronaut nutrition, and microbial studies. I suggest a **High-Resolution Camera** for a more detailed image and video to help inspect food prep areas and contamination areas and I suggest having 2 of them to mitigate the cons of only seeing a small area. The **Color Stereo Camera** provides 3D visuals of experiments

and food systems. The **Radiation Sensor** monitors radiation to prevent food/nutrient degradation. I also suggest the **Advanced Resistive Exercise Device (ARED)** to keep our astronauts healthy and strong and it would also boost productivity, safety, and mental health.

Why Not The Other Cards:

Low-Resolution Camera - Lacks detail needed for food system inspections or research.

Medium-Resolution Camera - Not specialized enough for bio/food monitoring

Infrared Camera - Detects heat but doesn't directly support food quality, growth, or storage needs.

Heimdall Camera - Too low in resolution and return (1) to justify inclusion.

Mass Spectrometer - Designed for atmosphere not for food-focused missions.

Magnetometer - Not relevant to food systems or crew survival.

Portable X-Ray Fluorescence Spectrometer - Useful for geology but not relevant to food quality, nutrition, or biology.

Sample Collection Device - Primarily used for collecting rock samples and more for geological science or planetary surface composition studies. It doesn't contribute directly to food safety, astronaut nutrition, environmental monitoring for food systems and our sustainability component goal is providing nutritious and delicious food. We already have the life science laboratory which is a great pick already.